

Quantum Feature Selection — Detailed Notes (Session 11)

Course: CS490/5590 — Quantum Computing Applications in Data Science, AI, & Deep Learning

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0) Introduction

- This lecture explains how to select a small, informative subset of features so quantum pipelines remain feasible on near-term processors.
 - A data point is $x \in \mathbb{R}^p$, where p is the number of original features and $\mathbb{R}^p$ is the set of p -tuples of real numbers.
 - A label is $y \in \{-1, +1\}$ for binary classification, where $\{-1, +1\}$ is the set of the two allowed class values.
 - A selected feature set is $S \subseteq \{1, \dots, p\}$ with $|S| = k$, where $|S|$ denotes the number of indices in S and k is the number of kept features.
 - A quantum feature map is $U_\varphi(x_S)$, where $U_\varphi(x_S)$ is a unitary circuit and $|\varphi(x_S)\rangle = U_\varphi(x_S) |0\rangle^{\otimes k}$ is the prepared state on k qubits.
- The guiding analogy is an orchestra on a tight stage. More instruments can create a richer sound, but each one must be tuned and kept in time. Each selected feature becomes a qubit to control. Feature selection keeps the instruments that carry the melody so the performance remains clean and achievable.

0.5) Road-map

- Recap quantum kernels and QSVMs.
- Explain why feature selection is necessary in quantum pipelines.
- Introduce classical and quantum mutual information as relevance scores.
- Show how to encode features and labels into quantum states.
- Derive practical estimators for quantum mutual information using swap tests.
- Present a kernel-based alternative to scoring features.
- Assemble a hybrid quantum–classical selection loop.
- Prototype on the breast-cancer dataset in Qiskit.
- Analyze noise sources and mitigation.
- Work through mini-exercises, an FAQ, and a summary.

1) Quick recap — quantum kernels and QSVMs

- A feature map circuit is $U_\varphi(x)$, where $U_\varphi(x)$ is a unitary built from single-qubit rotations and entanglers with angles determined by the real vector x .
- A quantum feature state is $|\varphi(x)\rangle = U_\varphi(x) |0\rangle^{\otimes n}$, where n is the number of qubits and $|0\rangle^{\otimes n}$ is the all-zero basis state.
- A quantum kernel is $K(x, x') = |\langle \varphi(x) | \varphi(x') \rangle|^2$, where $\langle \varphi(x) | \varphi(x') \rangle$ is the complex inner product and $|\cdot|^2$ is the squared magnitude that yields a real value in $[0, 1]$.
- A QSVM trains a classical SVM using the kernel matrix $K_{ij} = K(x_i, x_j)$, where K_{ij} is the similarity between training inputs x_i and x_j .
- The central idea is to move the burden of nonlinearity into the state preparation and then let a simple classical learner form the decision boundary using only similarities.

2) Why feature selection in quantum machine learning?

- Modern devices provide only a limited number of high-fidelity qubits. When each kept feature is mapped to one qubit, the hardware budget directly limits the number of features.
 - If each feature becomes a qubit, then the number of selected features k must be less than or equal to the number of available qubits, where both are positive integers.
- Deeper and wider circuits accumulate more gate error and decoherence. Reducing features shortens entangling patterns and lowers error rates.
 - Two-qubit gates are particularly error-prone, and they dominate depth in most feature maps.
- Kernel estimation requires evaluating $K(x_i, x_j)$ for many pairs. If there are N training points, then the number of distinct pairs scales on the order of N^2 . Reducing features lowers the cost per pair by reducing circuit depth and error mitigation overhead.
 - The symbol N denotes the number of training samples and is a positive integer.
- The practical analogy is a photographer choosing lenses for a long hike. Each lens adds weight and slows progress. A telephoto lens may be essential for wildlife, while two slightly different wide-angle lenses may be redundant. Feature selection is the decision to pack only the lenses that capture the scene well under the weight limit.

3) Mutual information as a relevance score

3.1 Classical mutual information

- Classical mutual information measures how much knowing one variable reduces uncertainty about another.
 - Let X and Y be random variables, where X takes values x and Y takes values y .
 - Let $p_{XY}(x, y)$ be the joint probability of seeing x with y . Let $p_X(x)$ and $p_Y(y)$ be the marginal probabilities.
 - The classical mutual information is

$$I(X; Y) = \sum_{x, y} p_{XY}(x, y) \log \left(\frac{p_{XY}(x, y)}{p_X(x) p_Y(y)} \right),$$

where the sum runs over all pairs (x, y) and the logarithm is taken in a chosen base so that the result is measured in bits or nats.

- A large value $I(X; Y)$ indicates that the variable X is informative about Y . This makes $I(\text{feature}; Y)$ a natural relevance score for selection.

3.2 Quantum mutual information

- Quantum mutual information measures total correlation between subsystems of a quantum state, including classical and entanglement correlations.
 - Let ρ_{XY} be a density matrix on a composite system, where ρ_{XY} is positive and has trace one.
 - The reduced states are $\rho_X = \text{Tr}_Y(\rho_{XY})$ and $\rho_Y = \text{Tr}_X(\rho_{XY})$, where Tr_Y and Tr_X denote partial traces.
 - The von Neumann entropy is $S(\rho) = -\text{Tr}(\rho \log \rho)$, where $S(\rho)$ is a nonnegative real number.
 - The quantum mutual information is

$$I_Q(X : Y) = S(\rho_X) + S(\rho_Y) - S(\rho_{XY}).$$

- A large value $I_Q(X : Y)$ indicates that the two subsystems carry strong shared structure. This is useful when the feature qubit and the label qubit share non-linear or entangled dependencies that classical linear measures would miss.
- The right mental model is two instruments playing in harmony. If each instrument alone sounds simple but together they create a strong pattern, then quantum mutual information will capture that pattern.

4) Encoding features and labels into quantum states

- The straightforward choice is to assign one qubit to each selected feature.
 - If the selected set has k features, then the prepared state uses k qubits for the features and optionally one more qubit for the label.
- A feature map circuit is $U_\varphi(x_S)$, where x_S is the vector of the selected components of x .
 - A common choice is a ZZFeatureMap with single-qubit $R_Z(\phi)$ rotations and two-qubit $R_{ZZ}(\phi)$ entanglers, where $R_Z(\phi) = e^{-i\phi Z/2}$ and $R_{ZZ}(\phi) = e^{-i\phi Z \otimes Z/2}$.
 - The symbol ϕ denotes a real-valued rotation angle measured in radians and the matrices Z and $Z \otimes Z$ are Pauli- Z and the tensor product of Pauli- Z with itself.
- There are two label-handling options.
 - A quantum label uses a dedicated label qubit $|y\rangle$, where $|y\rangle$ is prepared from the classical label.
 - A classical label uses conditional preparation, where runs are separated by the value of y and the results are combined in post-processing.
- The encoding rule should be clear and consistent so that a specific feature always maps to a specific qubit and a specific angle.

5) Estimating quantum mutual information with swap tests

5.1 Purity and the Rényi-2 proxy

- Directly estimating von Neumann entropy is challenging, so a practical proxy uses the Rényi-2 entropy, which is $S_2(\rho) = -\log \text{Tr}(\rho^2)$.
 - The symbol $\text{Tr}(\rho^2)$ is the purity of the state ρ and lies in the interval $[0, 1]$.
 - A larger purity indicates a more concentrated state, while a smaller purity indicates a more mixed state.
- The swap test estimates $\text{Tr}(\rho^2)$ by interfering two identical copies of the state.
 - The circuit prepares two registers in state ρ , adds a single-qubit ancilla, applies a controlled-SWAP between the registers, and measures the ancilla.
 - The probability of measuring the ancilla in the zero state is $p_0 = \frac{1}{2} + \frac{1}{2}\text{Tr}(\rho^2)$, so $\text{Tr}(\rho^2) = 2p_0 - 1$.

5.2 QMI workflow for one feature

- Prepare a dataset superposition for the feature qubit and the label qubit.
 - The joint state is $|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{k=1}^N |x_{k,i}\rangle |y_k\rangle$, where N is the number of samples, $|x_{k,i}\rangle$ encodes the i -th feature of sample k , and $|y_k\rangle$ encodes the label of sample k .
- Reduce this joint state to the feature qubit, the label qubit, and the two-qubit joint system.
 - The reduced states are ρ_i for the feature, ρ_ℓ for the label, and $\rho_{i\ell}$ for the pair.
- Estimate purities with swap tests.
 - The three values are $\text{Tr}(\rho_i^2)$, $\text{Tr}(\rho_\ell^2)$, and $\text{Tr}(\rho_{i\ell}^2)$.
- Compute the proxy quantum mutual information.
 - The final score is $I_Q(f_i : Y) \approx S_2(\rho_i) + S_2(\rho_\ell) - S_2(\rho_{i\ell})$, where each S_2 is computed from the corresponding purity.
- Rank features by this score from largest to smallest. The top entries are the most informative under the chosen map.
- The cost with respect to dataset size N is favorable because the shot complexity is governed by the number of qubits involved in each purity estimate rather than by N . The dataset contributes through how the state is prepared, not through repeated loops over samples.

6) Kernel-based feature scoring

- A practical alternative uses kernel accuracy differences when a single feature is removed.
 - Compute a baseline accuracy Acc_{all} with all selected features.
 - For each feature f_i , recompute accuracy as $\text{Acc}_{\text{all} \setminus i}$ after removing f_i .
 - Define the score as $s_i = \text{Acc}_{\text{all}} - \text{Acc}_{\text{all} \setminus i}$, where larger s_i indicates greater importance.
- This method is effective when a kernel pipeline already exists, since it avoids additional circuits for entropy estimation and reuses the same overlap-style estimators that were already implemented.
- The intuition is a taste test where one ingredient is removed at a time. If the dish noticeably worsens when an ingredient is removed, then that ingredient is important.

7) A hybrid quantum–classical selection loop

- The loop computes a relevance score for each candidate feature, ranks the features, and trains a model on the top k .
 - Initialize the set of candidate indices as $\{1, \dots, p\}$.
 - For each index i , compute a score $I_Q(f_i : Y)$ using the swap-test proxy or compute s_i using the kernel leave-one-out method.
 - Sort the pairs (i, score_i) by the score in descending order and select the first k indices.
 - Train either a QSVM with a feature map on the kept features or a classical model on the reduced matrix.
- This loop separates quantum data collection from classical decision making in a way that is robust and easy to iterate. The quantum device supplies high-value statistics, and the classical side performs ranking and validation.

8) Qiskit mini-experiment on the breast-cancer dataset

- The dataset supplies feature vectors $X \in \mathbb{R}^{N \times n}$ and labels $y \in \{-1, +1\}^N$. The goal is to keep a small set of features that preserve classification accuracy.
 - Normalize features, split into training and test sets, and choose an initial score such as variance to verify the flow.
 - Replace the placeholder score with the swap-test proxy for quantum mutual information once the pipeline is in place.
 - Instantiate a feature map such as a ZZFeatureMap with the number of qubits equal to the number of selected features.
 - Build a `QuantumKernel` to estimate the kernel entries and train an `SVC` on the precomputed kernel matrix.
- This experiment demonstrates the complete path from selection to training and shows how accuracy varies as the number of features changes.

9) Noise analysis and mitigation

- State-preparation and readout errors bias estimated probabilities and therefore bias purity and kernel values. Calibrating readout and inverting the calibration matrices reduces bias.
 - A calibration matrix is a 2×2 real matrix per qubit that describes measured bit-flip rates and can be inverted to correct observed counts.
- Decoherence reduces off-diagonal elements of density matrices and shrinks overlaps and purities. Techniques such as dynamical decoupling and shallower circuits help preserve coherence.
 - Reducing the number of two-qubit gates is an effective way to reduce decoherence impact.
- Shot noise introduces variance that makes rankings unstable. Adaptive shot allocation that assigns more shots to high-variance estimates stabilizes the ranking at the same total budget.
 - The number of shots is the number of repeated measurements per circuit and is a positive integer.
- Gate errors add systematic bias to overlaps. Zero-noise extrapolation and learned debiasing methods such as SKQD can reduce these effects.
 - Zero-noise extrapolation varies circuit depth to infer a zero-noise estimate by extrapolating measured values back to the zero-noise limit.

10) Mini-exercises

- Derive the swap-test probability formula for purity estimation and show that the ancilla-zero probability satisfies $p_0 = \frac{1}{2} + \frac{1}{2}\text{Tr}(\rho^2)$.
 - Explain how this implies $\text{Tr}(\rho^2) = 2p_0 - 1$ and show how to compute $S_2(\rho)$ from this value.
- Suppose there are four features and a single label qubit. Determine the number of physical qubits required to estimate quantum mutual information for one feature at a time using swap tests.
 - Account for the two copies required by the swap test and a single ancilla for control.
- Implement the kernel leave-one-feature-out score and measure the difference in accuracy between all features and $k = 2$ features on the iris binary subset.
 - Report the two accuracies and the scores for the removed features.
- Under two-qubit depolarizing error with rate $p = 0.02$ per entangler, estimate the shrinkage factor for purity when there are three entanglers in the path and discuss the bias this introduces in the quantum mutual information.
 - Use a simple model where purity scales by $(1 - \frac{4p}{3})^g$, where g is the number of entanglers.
- Explain why quantum mutual information can detect non-linear dependencies that Pearson correlation misses by analyzing an XOR-style dependence between a feature and the label.

11) Frequently asked questions

- It is possible to encode more than one feature per qubit by using data re-uploading, in which case feature selection chooses which parameters to keep rather than which qubits to keep.
- A swap test requires two copies of the state to estimate the purity, but classical-shadow estimators can approximate purity with fewer qubits and different trade-offs.
- Quantum mutual information is not always better than classical measures. It is most helpful when the relevant dependence involves phase relationships or entanglement that are not well captured by classical linear statistics.
- Feature selection reduces the number of qubits and the per-pair circuit depth, but it does not change the fact that kernel estimation for N points requires on the order of N^2 pairwise evaluations.

12) Summary

- The objective is to choose a compact set of features that preserves predictive power while fitting the realities of near-term hardware.
 - Quantum mutual information provides a principled score by comparing the entropy of a feature qubit, the entropy of a label qubit, and the entropy of their joint system.
 - Kernel-based leave-one-feature-out scoring provides a practical alternative that reuses the same circuits already built for training.
 - A hybrid loop separates quantum scoring from classical ranking and model training, which makes the process robust and easy to iterate.
- The conceptual picture is to keep the instruments that carry the melody, reduce the size of the orchestra to fit the stage, and deliver a performance that is both expressive and reliable.

MIQUBO: QUBO Representation of Feature Selection

D-Wave systems solve binary quadratic models (BQM)—the Ising model traditionally used in statistical mechanics and its computer-science equivalent, the quadratic unconstrained binary optimization (QUBO) problem. Given N variables $x_1, \dots, x_N$, where each variable x_i can have binary values 0 or 1, the system finds assignments of values that minimize,

$$\sum_i^N q_i x_i + \sum_{i < j}^N q_{i,j} x_i x_j,$$

where q_i and $q_{i,j}$ are configurable (linear and quadratic) coefficients. To formulate a problem for the D-Wave system is to program q_i and $q_{i,j}$ so that assignments of $x_1, \dots, x_N$ also represent solutions to the problem.

For feature selection, the Mutual Information QUBO (MIQUBO) method formulates a QUBO based on the approximation above for $I(X_k; Y)$, which can be submitted to the D-Wave quantum computer for solution.

The reduction of scope to permutations of three variables in this approximate formulation for MI-based optimal feature selection makes it a natural fit for reformulation as a QUBO:

	Optimization	Linear Terms	Quadratic Terms	Formula
Feature Selection	Maximize	$I(X_i; Y)$	$I(X_j; Y X_i)$	$\sum_{i \in S} \{I(X_i; Y) + \sum_{j \in S, j \neq i} I(X_j; Y X_i)\}$
QUBO	Minimize	$q_i x_i$	$q_{i,j} x_i x_j$	$\sum_i^N q_i x_i + \sum_{i < j}^N q_{i,j} x_i x_j$

You can represent each choice of $\binom{n}{k}$ features as the value of solution $x_1, \dots, x_N$ by encoding $x_i = 1$ if feature X_i should be selected and $x_i = 0$ if not. With solutions encoded this way, you can represent the QUBO in matrix format, $\mathbf{x}^T \mathbf{Q} \mathbf{x}$, where $\mathbf{Q}$ is an $n \times n$ matrix and $\mathbf{x}$ is an $n \times 1$ matrix (a vector) that should have k ones representing the selected features.

To map the feature-selection formula to a QUBO, set the elements of $\mathbf{Q}$ such that

- diagonal elements (linear coefficients) represent MI: $Q_{ii} \leftarrow -I(X_i; Y)$
- non-diagonal elements (quadratic elements) represent CMI: $Q_{ij} \leftarrow -I(X_j; Y | X_i)$

These QUBO terms are negative because the quantum computer seeks to minimize the programmed problem while the feature-selection formula maximizes. The following subsection codes this and then completes the formulation by adding the $\binom{n}{k}$ constraint to the QUBO.